



**LESSONS LEARNED
MADES**

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INTRODUCTION

The purpose of the lessons learned document for the MADES Project (a Progressive Web Application for inventory management) is to capture the project's lessons learned in a formal document for use by other project managers on similar future projects within Altamira Developers. This document may be used as part of new project planning for similar software development and PWA projects in order to determine what problems occurred—such as challenges with offline/online data synchronization, Service Workers, or conflict management in hybrid architectures—and how those problems were handled and may be avoided in the future. Additionally, this document details what went well with the project and why, so that other project managers may capitalize on these actions when implementing technologies like Angular, Spring Boot, and PostgreSQL. Project managers may also use this document to determine who the project team members were in order to solicit feedback for planning their projects in the future. This document will be formally communicated with the organization and will become a part of the organizational assets and archives.

LESSONS LEARNED APPROACH

The lessons learned from the MADES Project are compiled from project journal entries and sprint retrospectives throughout the project lifecycle. Lessons learned were also gathered from both realized and unrealized risks in the project risk register, software issue trackers, as well as through interviews with the Altamira Developers project team members and other stakeholders as necessary. The lessons learned from this project are to be used as references for future software development and PWA projects, and contain an adequate level of detail so that other project managers may have enough information on which to help base their project plans. The lessons learned in this document are categorized by project knowledge area. These knowledge areas consist of: procurement management, risk management, integration management, quality management, time management, cost management, scope management, human resource management, and communications management. NOTE: some knowledge areas may not contain lessons learned if none were documented throughout the project lifecycle.

LESSONS LEARNED FROM THIS PROJECT

The following chart lists the lessons learned for the MADES project. These lessons are categorized by project knowledge area, and descriptions, impacts, and recommendations are provided for consideration on similar future software development and Progressive Web App (PWA) projects. It is important to note that not only failures or shortcomings are included, but successes and best practices as well.



Category	Issue Name	Problem/Success	Impact	Recommendation
Scope Management	Flexible MVP Prioritization	Success - The team strictly prioritized core MVP features and successfully deferred advanced features (e.g., advanced multi-site support, automatic barcode scanning) to future versions.	Allowed the team to deliver a functional application quickly, enabling the business to transition away from manual inventory handling without delaying the launch.	Always establish a clear "NOT list" and flexible scope prioritization (using tools like the Trade-off slider) early in the project to ensure the MVP is delivered on time.
Quality Management	Offline/Online Synchronization Conflicts	Problem - Initial testing revealed conflicts in inventory discounting when synchronizing offline data with the server once connectivity was restored.	Caused a slight delay during the testing phase to refactor the Service Worker and backend logic, but it prevented severe financial impacts and inventory errors in production.	Always allocate extra testing time and technical spikes specifically for offline/online synchronization edge cases in PWA architectures. Implement robust conflict resolution strategies early.
Cost Management	Open-Source Technology Stack	The decision to exclusively use open-source technologies (Angular, Spring Boot, PostgreSQL, Docker) was strictly followed.	Kept licensing costs at zero and monthly server costs very low (under \$25/month for a VPS), successfully meeting the "Controlled Cost" strategic priority.	Continue evaluating and leveraging enterprise-grade open-source tools for similar projects to maintain low operational costs without sacrificing quality or scalability.



Risk Management	Service Worker Complexity	Problem - The complexity of configuring the PWA Service Workers to prevent offline data loss was underestimated during the initial planning.	Development velocity was slower than expected during the first two sprints, causing a slight bottleneck for the frontend team.	When adopting PWA technologies, ensure the development team includes technical spikes in the early sprints to accurately estimate caching complexity and mitigate technical risks early.
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LESSONS LEARNED KNOWLEDGE BASE / DATABASE

The lessons learned for the MADES Project will be contained in the organizational lessons learned knowledge base maintained by the project management office (PMO) at Altamira Developers. This information will be cataloged under the project's year (2026) and the type of project (Software Development / Progressive Web Application) for future reference. This information will be valuable for any project manager or development team assigned to a new software or PWA project in the future.

LESSONS LEARNED APPLIED FROM PREVIOUS PROJECTS

The MADES Project utilized several lessons learned from past projects:

1. Environment Standardization (Docker): The decision to containerize the entire development environment using Docker from day one was determined during the planning process by consulting the lessons learned from the Retail Web Portal Project of 2024. In that project, severe deployment delays occurred due to configuration mismatches between local environments and the production server ("it works on my machine" issues). Applying this lesson ensured seamless deployment for MADES.
2. Automated Quality Standards: The planning of acceptable code quality and automated testing standards was based on lessons learned from the Logistics Mobile App Project of 2025. By establishing minimum unit test coverage metrics and strict code linting rules early on, the project team was able to avoid major regression bugs and significantly reduce time spent on manual QA cycles.



PROCESS IMPROVEMENT RECOMMENDATIONS

As indicated in the lessons learned chart above, the MADES Project initially underestimated the complexity and testing requirements for offline/online data synchronization and Service Worker configurations. Not only is this a lesson learned for similar future Progressive Web App (PWA) projects; but Altamira Developers must ensure that all project managers and technical leads are aware of the need for specialized technical spikes to be included in the planning of all future projects utilizing new architectures. Therefore, it is recommended that prior to work beginning on any new PWA or hybrid project, the project manager must establish a mandatory architecture validation phase (technical spike) during the initial sprints to mitigate technological risks, and thoroughly brief the project sponsor on any potential impacts to the schedule.

